

SENTINEL-E: Anytime-Valid Sequential Detection for Streaming Visual Anomaly Surveillance

Quang-Vinh Dang 

British University Vietnam, Hung Yen, Vietnam

`vinh.dq4@buv.edu.vn`

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Abstract

Streaming visual-anomaly systems — illegal dumping among them — are trained and evaluated as fixed-sample-size classifiers: given a short clip, emit one decision and report clip-level precision and recall. Deployment does not look like that: a municipal camera runs for weeks, and what an operator manages is the number of false alarms accumulated over the whole monitoring period, because each one costs a dispatch. Alerting the first time a per-frame threshold is crossed on a continuous stream is optional stopping, and it inflates the true false-alarm rate by an amount clip-level evaluation cannot show: a per-frame rule fed p-values individually valid at the 1% level raises at least one false alarm on 99.8% of 40-minute null streams. Our contribution is an algorithm. Every anytime-valid sequential change detector in use — CUSUM, Shiryaev–Roberts, e-detectors, non-partitioned e-detectors — models a switch that, once it happens, lasts forever. A dumping event is an *episode*: it begins, flickers under intermittent occlusion, ends, and recurs weeks later at the same site. We replace the mixture over changepoint locations with a mixture over episodes, realised as a two-state Markov chain whose forward recursion collapses a sum over 2^t trajectories into two numbers. The resulting *episodic e-process* is an exact non-negative supermartingale — so Ville’s inequality gives time-uniform false-alarm control with no new machinery — costs $O(1)$ per frame against $O(t)$ for a general mixture over active starts, and contains the changepoint mixture exactly as the boundary case in which episodes never end, a strict generalisation along an axis the anytime-valid e-process family has left fixed at infinity. Around this core sits a per-camera conformal layer calibrated conditionally on observable context and corrected by exact Beta order-statistic levels, and a fleet layer in which a graph network steers each camera’s episode hazard through a *predictable* modulation — provably unable to damage the guarantee — combined across cameras by e-BH under arbitrary dependence. Two evidence sources back these claims. Large-horizon simulated streams (10^5 – 10^6 frames) show every classical and parametric baseline we test — fixed threshold, per-frame p-value threshold, CUSUM, Shiryaev–Roberts, a parametric e-detector, E-SHIFT — failing to hold its nominal false-alarm level, while SENTINEL-E and the changepoint mixture it strictly contains both hold it; a paired comparison on a shared betting grid resolves a real advantage for the episodic mixture at two and four recurring episodes and a real, smaller cost at one. We also report where the guarantee stops holding: an unmodelled covariate shift beyond $\times 3.0$ inflates the rate unless a shift-weighting safeguard is engaged, at a steep cost in power. New in this revision: the pipeline is also run, unmodified, on real surveillance video (CUHK Avenue and UCSD Ped2) with a real optical-flow backbone rather than a simulator. This surfaces a limitation the simulated streams could not: real per-frame scores are autocorrelated across hundreds of frames (207 on Avenue, 357 on Ped2), close to or beyond a real episode’s own

duration, leaving only a narrow lag window in which validity and power coexist at all. Inside that window the guarantee holds on both real corpora — realised false-alarm rate 0.000 on Ped2 and 0.020 on Avenue at $\alpha = 0.1$ — while every classical baseline tested fails to hold it on the same real streams; detection power is real but modest on Ped2 and weak on Avenue, whose real episodes are shorter. This gap, not the simulated results, is the paper’s central open problem. SENTINEL-E costs 0.61 μ s per frame, 0.0016% of the detector it wraps.

Keywords: video anomaly detection; illegal dumping detection; anytime-valid inference; e-values; conformal prediction; sequential change detection; false discovery rate; video surveillance; edge deployment

Declarations

Funding. No funding was received for this work.

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Author contributions. The sole author conceived the method, implemented the library and experiments, and wrote the manuscript.

Data and code availability. All code, experiment scripts and generated results are available in the project repository, to be disclosed upon acceptance to preserve the double-anonymous review process; a private link is available to editors and reviewers on request. The corpora referenced in the manuscript’s Related Work (CUHK Avenue, UCSD Ped2, TACO, ZeroWaste, UAVVaste, AerialWaste, MARIDA, MADOS, Mivia-IWDD) are distributed by their respective authors under their own terms; no data from them is redistributed here.

Ethical approval. Not applicable. No human-subjects data was collected for this work. The real video corpora used (CUHK Avenue, UCSD Ped2) are long-standing, publicly released video-anomaly-detection benchmarks distributed by their original authors for research use; this paper is a downstream, secondary use of that already-public release, not new data collection, and involves no interaction with human subjects. This work concerns the statistical calibration of alerting systems, not the governance of their deployment; the manuscript’s Limitations section gives a substantive treatment of the human-oversight, accountability, and disclosure questions that automated surveillance of public space raises, rather than treating the topic as out of scope.